

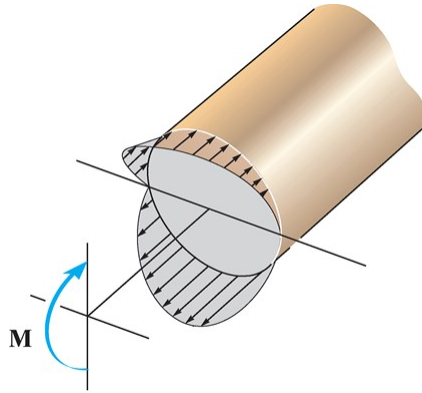
Solid Mechanics (EMCH260 Section 002, Fall 2021)

Mid-Term Exam 3, 8:30 AM, Friday, Nov. 19th 2021

Name (First/Last): _____ Score: _____/100'

1. Single choice questions. There is one and only one correct answer for each question. (10 × 3' = 30' in total)

Q1: What kind of stress is induced in the cross section by a bending moment?

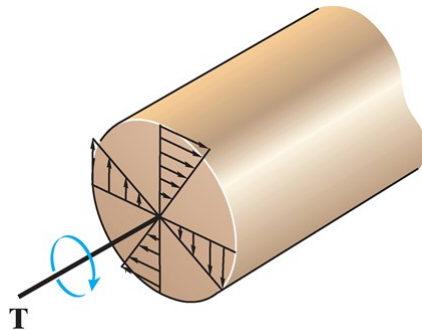


- A. Shear stress
- B. Normal stress

Answer: B

Explanation: According to the flexure formula, $\sigma = -\frac{My}{I}$, normal stress is induced in the cross section by a bending moment. See page #10 in the PPT Ch6.3 and also page #8 in the PPT Ch8.1.

Q2: What kind of stress is induced in the cross section by a torque?



- A. Shear stress
- B. Normal stress

Answer: A

Explanation: According to the torsion formula, $\tau = \frac{T\rho}{J}$, shear stress is induced in the cross section by a torque. See page #7 in the PPT Ch5.1 and also page #9 in the PPT Ch8.1.

Q3: For a thin-walled pressure vessel, when under internal pressure, what type of stress is induced in the circumferential (hoop) and longitudinal (axial) direction?



- A. Shear stress
- B. Normal stress

Answer: B

Explanation: Normal stress is induced in both circumferential (hoop) and longitudinal (axial) direction. See page #4 in the PPT Ch8.2.

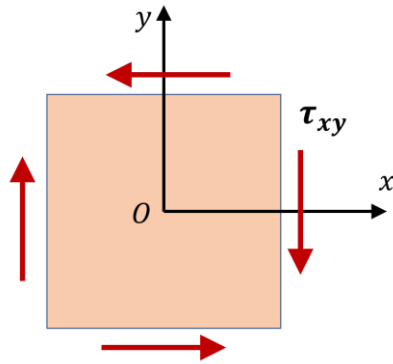
Q4: For a thin-walled pressure vessel, when under internal pressure (see the same picture as Q3), which stress is higher?

- A. The stress in the circumferential (hoop) direction
- B. The stress in the longitudinal (axial) direction

Answer: A

Explanation: Normal stress induced in circumferential (hoop) direction is 2 times of normal stress induced in longitudinal (axial) direction. See page #4 in the PPT Ch8.2.

Q5: For a plane stress state, the coordinate system Oxy is established as shown in the figure. If the shear stress τ_{xy} is plotted in this way, then, according to the **sign convention**, what is the sign of the shear stress τ_{xy} ?

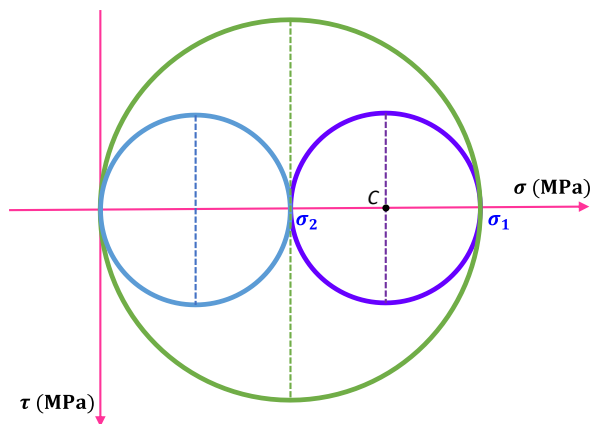


- A. Positive
- B. Negative

Answer: B

Explanation: A shear stress component is positive provided it acts in the positive coordinate direction on the positive face of the element. See page #11 in the PPT Ch9.1.

Q6: For a plane stress state, if the three Mohr's circles are drawn in the figure below, then, what is the absolute maximum shear stress ($\tau_{\max}^{\text{abs}}$)?

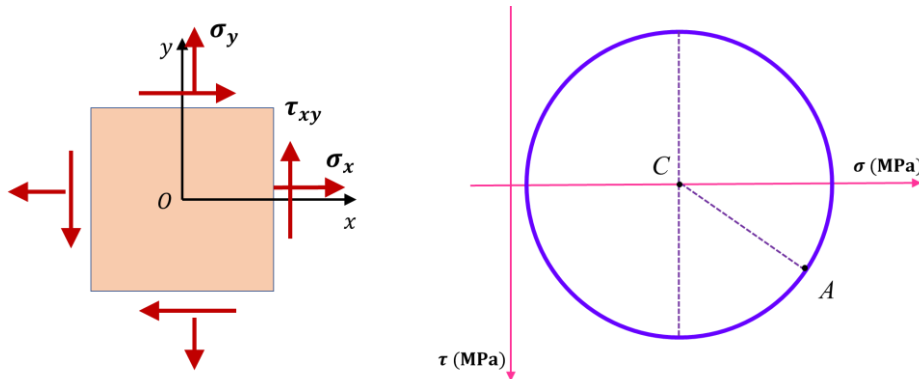


- A. $\sigma_1/2$
- B. $\frac{\sigma_1 - \sigma_2}{2}$

Answer: A

Explanation: The absolute maximum shear stress ($\tau_{\max}^{\text{abs}}$) is always read from the largest Mohr's circle. For the two plane stresses having same positive sign, the $\tau_{\max}^{\text{abs}}$ is then equal to half of σ_1 . See page #9 in the PPT Ch9.3.

Q7: For a plane stress state, if the three components of the stress are σ_x , σ_y , and τ_{xy} (left figure), then the reference point A in the Mohr's circle (right figure) always has stress component:

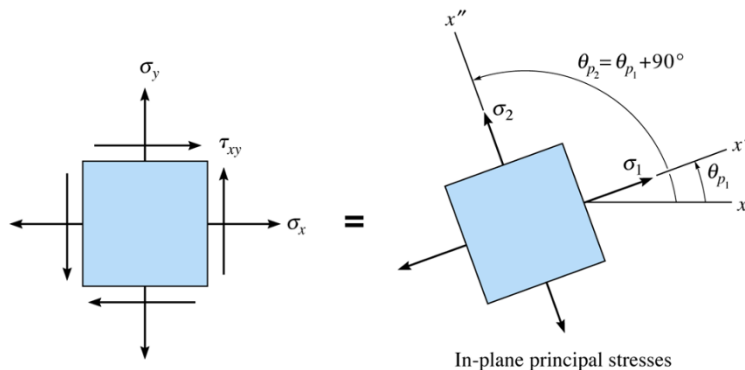


- A. $(\frac{\sigma_x + \sigma_y}{2}, \tau_{xy})$
- B. (σ_x, τ_{xy})

Answer: B

Explanation: We always use (σ_x, τ_{xy}) as reference point A to plot the Mohr's circle. See page #7 in the PPT Ch9.2.

Q8: The corresponding shear stress ($\tau_{x'y'}$) for the two in-plane principal stresses (σ_1 and σ_2 in the right figure) is always zero. Is this statement true or false?

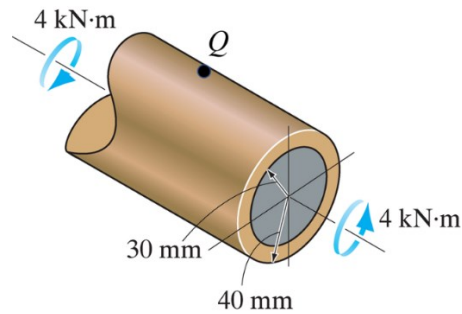


- A. True
- B. False

Answer: A

Explanation: There is no shear stress acting on the principal planes. See page #19 in the PPT Ch9.1. It can be also seen from the Mohr's circle.

Q9: The hollow circular shaft is subjected to the torque. For a surface point Q (see figure below), which statement is correct?

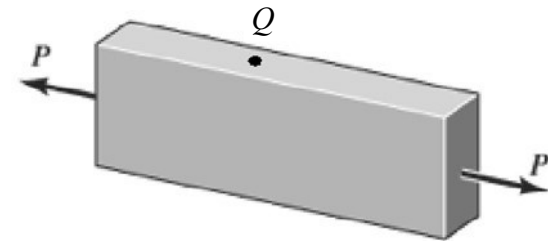


- A. The **stress** at the surface point Q is a **plane stress** problem
- B. The **strain** at the surface point Q is a **plane strain** problem

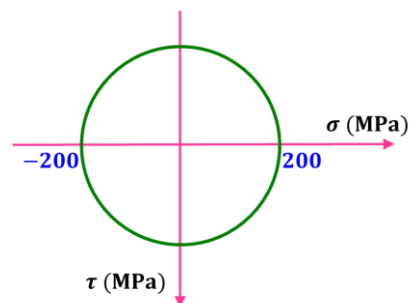
Answer: A

Explanation: For surface points, there is no shear stress in the surface plane and there is no normal stress in the direction perpendicular to the surface ($\sigma_z = 0$). Therefore, the stress at the surface point Q is a plane stress problem.

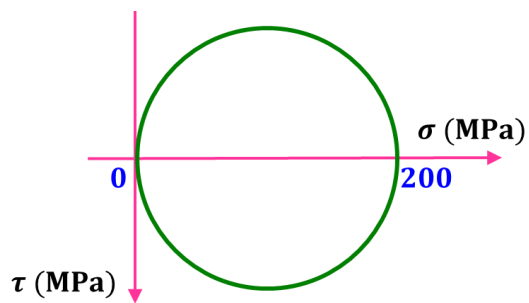
Q10: In the uniaxial tension, if the external force P induces normal (tensile) stress of 200 MPa in the cross section. Which Mohr's circle corresponds to the stress state for a surface point Q ?



A.



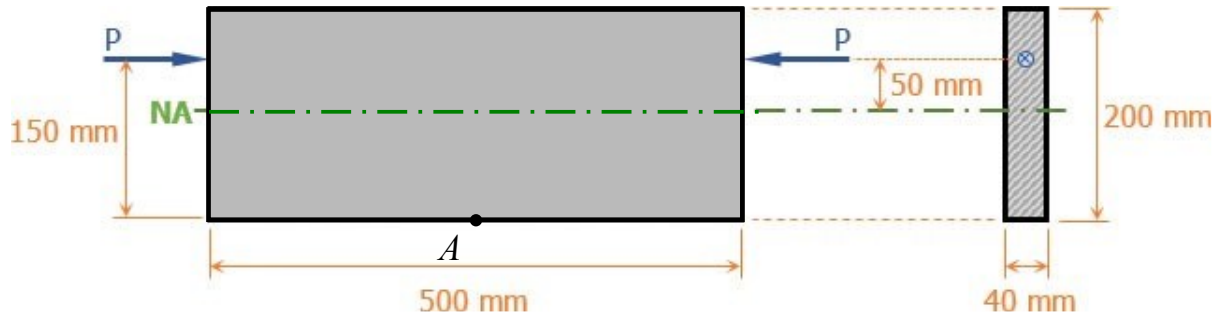
B.



Answer: B

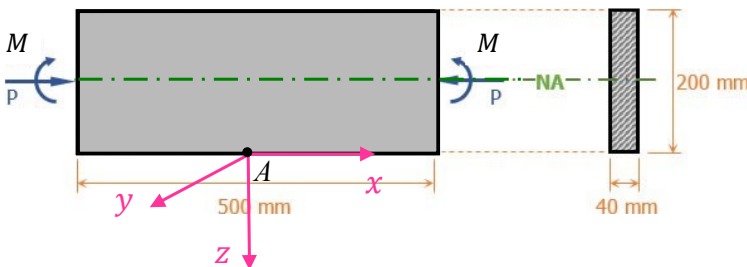
Explanation: In uniaxial tension, when the section is perpendicular to the axial direction of the object, there is only tensile (normal) stress acting on the plane. Therefore, the principal stresses are $\sigma_1 = \frac{P}{A} = 200 \text{ MPa}$, $\sigma_2 = 0$. Then, the Mohr's circle is the figure on the right. See page #4 in the PPT Ch9.4.

2. A cast iron link has dimension of $40 \times 200 \times 500$ mm as shown in the figure. The compressive load $P = 400$ kN is applied to the ends of the link along a longitudinal axis that is located 150 mm above the bottom of the link (50 mm above the neutral axis). Determine the principal stresses and the absolute maximum shear stress developed at the bottom surface point A . (30' in total)



Solutions:

The normal stress developed at the bottom surface point A is the combination of axial compressive force and bending stress. We can move the compressive force P to the central (longitudinal) axis and then put on the equivalent bending moment M . The equivalent figure is shown below.



The equivalent bending moment $M = 0.05 \times P = 0.05 \times 400 = 20$ kN · m.

For the bottom surface point A , establish xyz coordinate system as shown. Then, $\sigma_z = 0$, meaning that the stress state at point A is plane stress.

For the bottom part of the link (**tensile** stress region), the stress along x-direction is:

$$\sigma_x = \sigma_{\text{bending}} - \sigma_P$$

$$\sigma_x = \frac{M \times h/2}{\frac{1}{12}bh^3} - \frac{P}{A} = \frac{20000 \times \frac{0.2}{2}}{0.04 \times \frac{0.2^3}{12}} - \frac{400000}{0.04 \times 0.2} = 25 \text{ MPa (tensile)}$$

$$\sigma_y = 0$$

Since there is no shear stress component ($\tau_{xy} = 0$), we can immediately know the two normal stresses are the principal stresses (assign larger stress to σ_1):

$$\sigma_1 = \sigma_x = \boxed{25 \text{ MPa}}$$

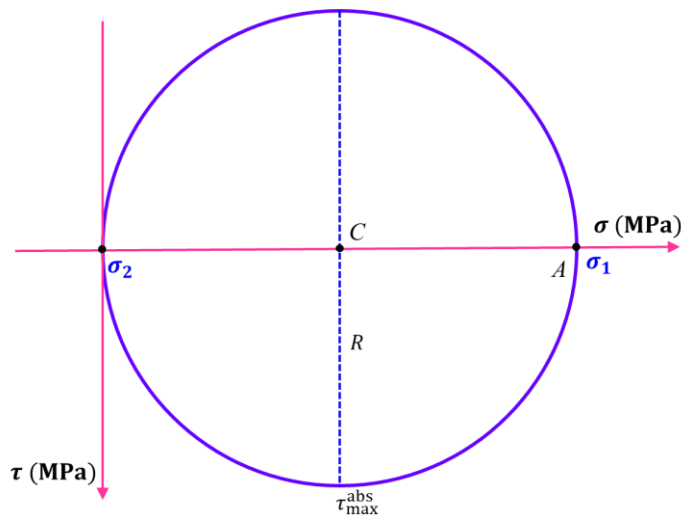
$$\sigma_2 = \sigma_y = \boxed{0 \text{ MPa}}$$

Based on the principal stresses, we can draw the Mohr's circle as shown in the figure.

The Mohr's circle is centered at $C\left(\frac{\sigma_1 + \sigma_2}{2}, 0\right) = C(12.5, 0)$ with radius $R = \frac{\sigma_1 - \sigma_2}{2} = 12.5$.

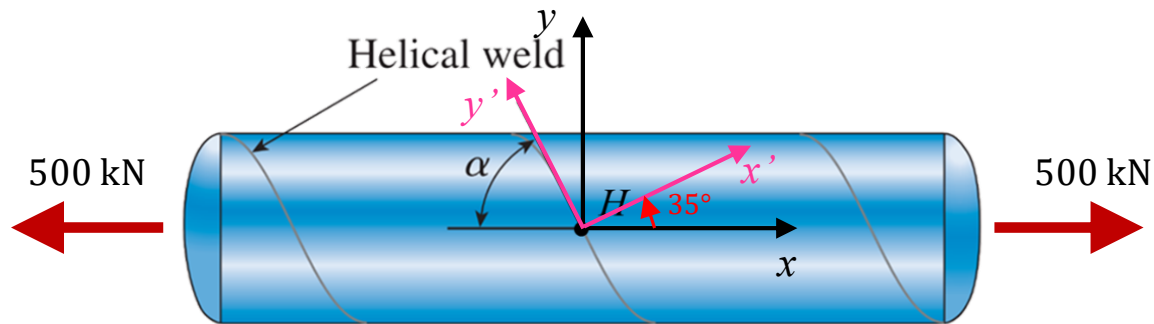
The absolute maximum shear stress can be obtained by the three Mohr's circles:

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = \boxed{12.5 \text{ MPa}}$$



3. A pressurized tank is constructed with a helical weld that makes an angle $\alpha = 55^\circ$ with the longitudinal axis. The tank has inner radius $r = 0.6$ m, wall thickness $t = 18$ mm, and internal pressure $p = 1.2$ MPa. The pressurized tank is also subjected to the tensile force of 500 kN shown in the figure. The tank has modulus of elasticity $E = 200$ GPa and Poisson's ratio $\nu = 0.25$. For a point H on the tank surface, determine the following quantities (40' in total):
- The principal stresses and absolute maximum shear stress. (10')
 - The normal and shear stresses acting on planes parallel and perpendicular to the weld; show these stresses on a properly oriented stress element. (10')
 - The circumferential (hoop) and longitudinal strains. (10')
 - The absolute maximum shear strain. (10')

Hint: the cross-section area of the tank can be calculated as $2\pi r t$.



Solutions:

Establish xyz -system as shown (z -direction is perpendicular to the surface). The problem is a plane stress (xy -plane) problem.

(a) Both the tensile force and inner pressure will induce normal (tensile) stress to the surface point.

For the tensile force, the resultant normal stress is:

$$(\sigma_x)_F = \frac{F}{A} = \frac{500 \times 1000}{2\pi \times 0.6 \times 0.018} = 7.368 \text{ MPa (tensile, along x direction)}$$

For the inner pressure, the circumferential (hoop) stress is

$$\sigma_{\text{hoop}} = \frac{pr}{t} = \frac{1.2 \times 10^6 \times 0.6}{0.018} = 40 \text{ MPa (tensile, along y direction)}$$

The longitudinal stress is

$$\sigma_{\text{long}} = \frac{pr}{2t} = \frac{1.2 \times 10^6 \times 0.6}{2 \times 0.018} = 20 \text{ MPa (tensile, along x direction)}$$

Then, the total normal stress in x -direction is:

$$\sigma_x = 20 + 7.368 = 27.368 \text{ MPa (tensile)}$$

The total normal stress in y -direction is:

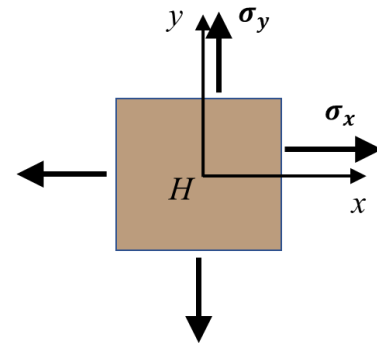
$$\sigma_y = \sigma_{\text{hoop}} = 40 \text{ MPa (tensile)}$$

The stress state in Hxy system is plotted. Both stress components are tensile.

Since there is no shear stress component ($\tau_{xy} = 0$), we can immediately know the two normal stresses are the principal stresses (assign larger stress to σ_1):

$$\sigma_1 = \sigma_y = \boxed{40 \text{ MPa}}$$

$$\sigma_2 = \sigma_x = \boxed{27.368 \text{ MPa}}$$



Based on the principal stresses, we can draw the Mohr's circle as shown in the figure.

The Mohr's circle is centered at $C\left(\frac{\sigma_1 + \sigma_2}{2}, 0\right) = C(33.684, 0)$ with radius $R = \frac{\sigma_1 - \sigma_2}{2} = 6.316$.

The absolute maximum shear stress can be obtained by the three Mohr's circles:

$$\tau_{\text{max}}^{\text{abs}} = \frac{\sigma_1}{2} = \boxed{20 \text{ MPa}}$$

(b) From the reference point $A(\sigma_x, \tau_{xy}) = A(27.368, 0)$ to the plane along which we want to know the stresses, we need to rotate the CA in the Mohr's circle by $2\theta = 2 \times (90^\circ - \alpha) = 2 \times 35^\circ = 70^\circ$ (counterclockwise).

Using the Mohr's circle, the new stresses correspond to the point P (see figure) are

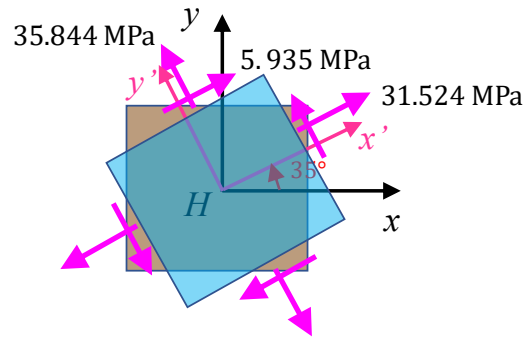
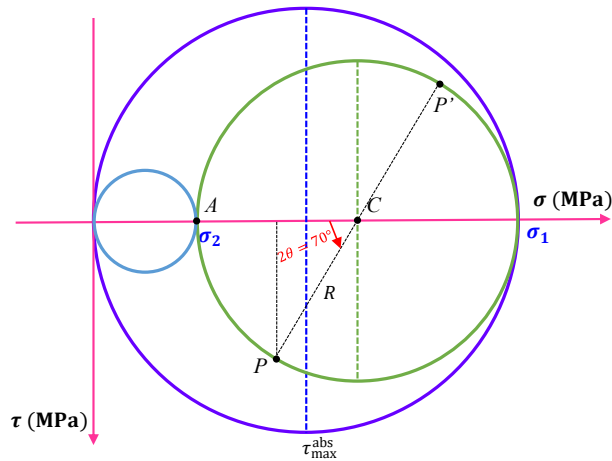
$$\sigma_{x'} = \sigma_P = \sigma_C - R \cos 2\theta = 33.684 - 6.316 \times \cos 70^\circ = \boxed{31.524 \text{ MPa}}$$

$$\tau_{x'y'} = \tau_P = R \sin 2\theta = 6.316 \times \sin 70^\circ = \boxed{5.935 \text{ MPa}}$$

$\sigma_{y'}$ can be regarded as continuing to rotate by another 90° , i.e. 180° on the Mohr's circle.

$$\sigma_{y'} = \sigma_{P'} = \sigma_C + R \cos 2\theta = 33.684 + 6.316 \times \cos 70^\circ = \boxed{35.844 \text{ MPa}}$$

The oriented stress element is shown by the dashed square with stresses labeled.



(c) For the xyz -coordinate system established, using the generalized 3D Hooke's law, the circumferential (hoop) strain (in y -direction) is

$$\varepsilon_y = \varepsilon_{\text{hoop}} = \frac{\sigma_y - \nu\sigma_x}{E} = \frac{40 \times 10^6 - 0.25 \times 27.368 \times 10^6}{200 \times 10^9} = \boxed{1.6579 \times 10^{-4}}$$

The longitudinal strain (in x -direction) is

$$\varepsilon_x = \varepsilon_{\text{long}} = \frac{\sigma_x - \nu\sigma_y}{E} = \frac{27.368 \times 10^6 - 0.25 \times 40 \times 10^6}{200 \times 10^9} = \boxed{8.684 \times 10^{-5}}$$

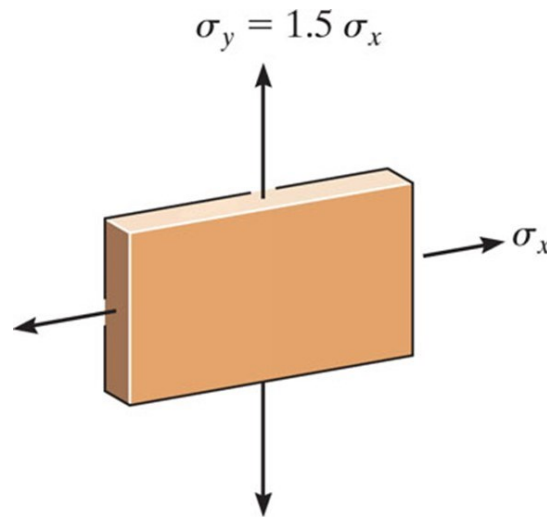
(d) The shear modulus can be obtained by the material property relationship

$$G = \frac{E}{2(1 + \nu)} = \frac{200 \times 10^9}{2 \times (1 + 0.25)} = 80 \text{ GPa}$$

Finally, the absolute maximum shear strain is obtained by using the maximum shear stress and generalized Hooke's law:

$$\gamma_{\text{max}}^{\text{abs}} = \frac{\tau_{\text{max}}^{\text{abs}}}{G} = \frac{20 \times 10^6}{80 \times 10^9} = \boxed{2.5 \times 10^{-4} \text{ rad}}$$

*****Bonus Problem***** For the state of stress shown in the figure, determine the maximum tensile stress σ_x that can be applied to the plate, if a tensile stress $\sigma_y = 1.5\sigma_x$ is also applied and the absolute maximum shear stress in the plate cannot exceed 75 MPa. (10' in total)



Solutions:

This is the plane stress problem. Given σ_x , $\sigma_y = 1.5\sigma_x$, and there is no shear stress, we can immediately know the two normal stresses are the principal stresses (assign larger stress to σ_1):

$$\sigma_1 = \sigma_y = 1.5\sigma_x$$

$$\sigma_2 = \sigma_x$$

We can then plot the Mohr's circle with center $C \left(\frac{\sigma_x + \sigma_y}{2}, 0 \right) = C(1.25\sigma_x, 0)$ and radius $R = \frac{\sigma_1 - \sigma_2}{2} = 0.25\sigma_x$.

According to the Mohr's circle, the absolute maximum shear stress is

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = 0.75\sigma_x = 75 \text{ MPa}$$

Solving this equation: $\sigma_x =$ **100 MPa**

